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Style

Truth and Nil: Niladic

Wuxia

俠

The Hero

U+4FA0

Nil



The empty set — quiet failure value; absorbs; Nil⊡function → Nil

U+2205

Truth

⊤

Verum — first-class truth; no true anymore

U+22A4

Truth and Nil: Monadic

Invert

~

Invert — monadic only (tilde). Scalar: ~ 0 → 1, ~ 1 → 0 (user wording to confirm). Unary scalar verbs operate on array back ranks (new 2026-10-11)

U+007E

Not

¬

Not — monadic boolean not: ¬ Truth yields Nil, ¬ Nil yields Truth (new 2026-10-14)

U+00AC

Congruent

≡

Ambivalent: monadic binds train to lhs⊡function

U+2261

Truth and Nil: Dyadic

Less Than



Less Than — dyadic < A B yields Truth when A < B, Nil otherwise (new 2026-10-14)

U+003C

Greater Than



Greater Than — dyadic > A B yields Truth when A > B, Nil otherwise (new 2026-10-14)

U+003E

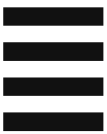
Congruent



Ambivalent: dyadic ≡ A B yields Truth when A and B have the same binding (same location in the system tables), Nil otherwise

U+2261

IsAt



IsAt — dyadic ≣ X Y yields Truth when X and Y share the same pointer address, Nil otherwise (new 2026-10-14)

U+2263

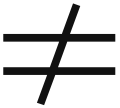
Equality



Equality — dyadic = A B yields Truth when A = B, Nil otherwise (new 2026-10-14)

U+003D

Unequal



Unequal — dyadic ≠ A B: sugar for ¬ = A B (boolean Not of Equality) (new 2026-10-14)

U+2260

Less Or Equal



Less Or Equal — dyadic ≤ A B yields Truth when A ≤ B, Nil otherwise (new 2026-10-14)

U+2264

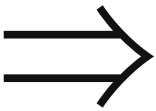
Greater Or Equal



Greater Or Equal — dyadic ≥ A B yields Truth when A ≥ B, Nil otherwise (new 2026-10-14)

U+2265

Implies



Implies — logical implication; ∅ ⇒ ∅ → ⊤

U+21D2

forall



Train quantifier — every element must evaluate non-nil. Idiom: dyadic NAND is ¬ ∀ 𝕀𝕀 X Y

U+2200

Approx



Approx — dyadic ≈ A B yields Truth when|A - B|is within 1e-6, Nil otherwise (new 2026-10-14)

U+2248

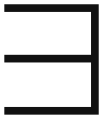
UnApprox



UnApprox — dyadic ≉ A B: sugar for ¬ ≈ A B (boolean Not of Approx) (new 2026-10-14)

U+2249

Forsome



Existential — at least one element evaluates non-nil

U+2203

Fornone



Not-forsome — defined as ~Forsome

U+2204

Arithmetic: Niladic

Signum / Plus



Ambivalent: Niladic + yields 0.
(Was ∓ U+2213, retired 2026-10-12

U+002B

Negate / Subtract

Niladic - yields 0 (new 2026-10-11

U+002D

Plus (Complex)



Plus (Complex), aka Sum Complex, Add Complex —; Niladic: sugar for ℿ 0 0, the complex zero (new 2026-10-14)

U+2A22

Times (Complex)



Times (Complex) —; Niladic: sugar for ℿ 1 0, the complex one (new 2026-10-14)

U+2A30

Minus (Complex)



Minus (Complex) —; Niladic: sugar for ⨿ 0 0, the complex zero (new 2026-10-14)

U+235B

Times (Polar)



Times (Polar) —; Niladic: the polar one, 2×1 [1; 0] (new 2026-10-14)

U+2A02

Divide (Complex)



Divide (Complex) —; Niladic: sugar for ⨿ 1 0, the complex one (new 2026-10-14)

U+2A2C

Divide (Polar)



Divide (Polar) —; Niladic: the polar one, 2×1 [1; 0] (new 2026-10-14)

U+2A38

Factorial

!

Factorial —; Niladic ! yields 1 (new 2026-10-12)

U+0021

Percent

%

Percent —; Niladic % is 0 (new 2026-10-12

U+0025

Square



Square — ⊠ X is X squared.; Niladic ⊠ yields 1, derived from Multiply (new 2026-10-12)

U+22A0

Multiply



*Multiply —; Niladic × yields empty array. Replaces Star * U+002A (new 2026-10-12; boxed forms 2026-10-12*

U+2A2F

Arithmetic: Monadic

Signum

+

Ambivalent: monadic Signum (less than zero yields −1, greater than zero yields 1, zero yields 0; back-rank scalar unary)

U+002B

Negate

-

Negate (monadic - X, unary minus)

U+002D

Conjugate

⊶

Conjugate — monadic ⊶ X: complex conjugate; negates the imaginary part, preserving orientation (on [r,i] yields [r,−i]; on [r;i] yields [r;−i]) (new 2026-10-15)

U+22B6

Double

+2

Double — monadic +⁼ X yields 2 × X (new 2026-10-15)

U+2A27

Sum

∑

Sum — sugar for reduce-addition: ∑ X means / + X (redefined 2026-10-12)

U+2211

Square Root

√

Square Root — monadic √ X yields the square root of X (new 2026-10-15)

U+221A

Plus (Complex)

⨁

Plus (Complex), aka Sum Complex, Add Complex —; Monadic: if X is scalar, sugar for ℙ X 0 (promote to complex); else identity, yields X.

U+2A22

Times (Complex)

⨂

Times (Complex) —; Monadic: if X is scalar, sugar for ℙ X 0 (promote to complex); else identity, yields X.

U+2A30

Minus (Complex)



Minus (Complex) —; Monadic: if X is scalar, sugar for ℚ X 0 (promote to complex); else identity, yields X.

U+235B

Times (Polar)



Times (Polar) —; Monadic: if X is scalar, promote to polar 2×1; else identity, yields X.

U+2A02

Divide (Complex)



Divide (Complex) —; Monadic: if X is scalar, sugar for ℚ X 0 (promote to complex); else identity, yields X.

U+2A2C

Divide (Polar)



Divide (Polar) —; Monadic: if X is scalar, promote to polar 2×1; else identity, yields X.

U+2A38

Percent

%

Percent —; monadic % X is absolute value, a back-rank operator.; monadic redefined 2026-10-12)

U+0025

Star

*

*Star — exponentiate: monadic * X is 1, because the missing exponent Y is assumed 0, giving X^0 (new 2026-10-12). Idiom: X-th root of Y is * Y ÷ 1 X*

U+002A

Quotient

÷

Quotient —; Monadic ÷ X: if X is a square array, yields the matrix inverse; else reciprocal, a back-rank scalar operator (new 2026-10-12)

U+00F7

Multiply

×

Multiply —; Monadic × X yields X when X numeric; on boxed X, catenates once.; monadic redefined 2026-10-12)

U+2A2F

Ceiling



Ceiling —; monadic ⌈ X rounds up
(from APL; new 2026-10-12)

U+2308

Floor



Floor —; monadic ⌊ X rounds down
(from APL; new 2026-10-12)

U+230A

Sine



Sine (new 2026-10-12)

U+25DD

Cosine



Cosine (new 2026-10-12)

U+25DE

Arccosine



Arccosine (new 2026-10-12)

U+25F6

Arcsine



Arcsine (new 2026-10-12)

U+25F7

Tangent



Tangent (new 2026-10-12)

U+2AD7

Arctangent



Arctangent (new 2026-10-12)

U+2AD8

Hyperbolic Sine



Hyperbolic Sine (new 2026-10-12)

U+2AA6

Hyperbolic Cosine



Hyperbolic Cosine (new 2026-10-12)

U+2AA7

Hyperbolic Arcsine



Hyperbolic Arcsine (new 2026-10-12)

U+2ABD

Hyperbolic Tangent



Hyperbolic Tangent (new 2026-10-12)

U+2AD1

Natural Log



Natural Log —; monadic ⍁ X is ln X.; Monadic ⍁ on a polar 2×1 yields the rectangular complex pair [ln r, θ].

U+237C

Antilog



Antilog — sugar for Star: ⍋̇ X is e raised to the X. Monadic ⍋̇ on a rectangular complex pair yields the polar 2×1 [e^a; b] (new 2026-10-12; glyph was 𝒆 U+1D58A until 2026-10-14)

U+29CA

Hyperbolic Arctangent



Hyperbolic Arctangent (new 2026-10-12)

U+2A4D

Hyperbolic Arccosine



Hyperbolic Arccosine (new 2026-10-12)

U+2ABE

Factorial

!

Factorial —; monadic ! X is X factorial

U+0021

Fourier



Fourier — monadic F X: FFT of X (new 2026-10-15)

U+03DC

InverseFourier



InverseFourier — monadic ⅁ X: inverse FFT of X (new 2026-10-15)

U+2132

Base Ten Log



Base Ten Log — ⍙ X is log base 10 of X, sugar for ⍦ 10 X (new 2026-10-12)

U+2A5F

Magnitude



Magnitude — monadic ⍫X: absolute value of X (new 2026-10-16)

U+2357

Arithmetic: Dyadic

Plus



Ambivalent: dyadic + X Y yields X + Y (addition).; dyadic 2026-10-15)

U+002B

Subtract

Subtract (dyadic - X Y yields X - Y, first minus second), the subtract that works with the / right-fold Reduce.; dyadic 2026-10-12)

U+002D

Plus (Complex)



Plus (Complex), aka Sum Complex, Add Complex —; dyadic: sum of complex numbers (1×2 rectangular), per the structural complex convention; if either or both X and Y are scalar, promote via ℿ before adding.

U+2A22

Times (Complex)



Times (Complex) —; dyadic: product of complex numbers, per the structural complex convention; if either or both X and Y are scalar, promote via ℿ before multiplying.

U+2A30

Minus (Complex)



Minus (Complex) —; dyadic: difference of complex numbers, per the structural complex convention; if either or both X and Y are scalar, promote via ℿ before subtracting.

U+235B

Times (Polar)



Times (Polar) —; dyadic: product of polar numbers; uses the 2×1 oriented vector to hold magnitude and direction: magnitudes multiply, angles add; if either or both X and Y are scalar, promote to polar 2×1 before multiplying.

U+2A02

Divide (Complex)



Divide (Complex) —; dyadic: quotient of complex numbers, per the structural complex convention; if either or both X and Y are scalar, promote via ℿ before dividing.

U+2A2C

Divide (Polar)



Divide (Polar) —; dyadic: quotient of polar numbers; uses the 2×1 oriented vector to hold magnitude and direction: magnitudes divide, angles subtract; if either or both X and Y are scalar, promote to polar 2×1 before dividing.

U+2A38

Percent

%

Percent —; dyadic % X Y yields the remainder of X divided by Y

U+0025

Star

*

*Star — exponentiate: dyadic * X Y is X to the power Y (left=base, right=exponent)*

U+002A

Quotient

÷

Quotient —; dyadic ÷ X Y divides X by Y, yielding a real; division by zero yields infinity, except 0/0 which yields Nil.

U+00F7

Multiply

×

Multiply —; dyadic multiplies its arguments; when X is boxed, × X Y catenates X repeated Y times.

U+2A2F

Factorial

!

Factorial —; dyadic ! X Y is the binomial coefficient Y choose X.

U+0021

Ceiling

⌈

Ceiling —; dyadic ⌈ X Y is maximum

U+2308

Floor

⌊

Floor —; dyadic ⌊ X Y is minimum

U+230A

Natural Log



Natural Log —; Dyadic ⍦ X Y is log base X of Y, sugar for ÷ ⍦ Y ⍦ X (new 2026-10-12)

U+237C

Square



Square — ⊠ X is X squared.; Dyadic ⊠ X Y squares X Y times.

U+22A0

Constants and Chance: Niladic

Half

½

Half —; niladic: 0

U+00BD

Pi

π

Pi — niladic π: value of pi (new 2026-10-14)

U+03C0

Bell

𝔅

Bell —; niladic: random point in (-1, 1), exclusive

U+1D56D

Real

ℜ

Real — niladic: random real number between 0 and 1; sugar for ½ + 1 𝔅 (new 2026-10-13)

U+1D57D

Gamma



Gamma — niladic γ: Euler-Mascheroni constant (new 2026-10-14)

U+03B3

Golden



Golden — niladic φ: golden mean (new 2026-10-14)

U+03C6

Euler



Euler — niladic ℇ: Euler's constant (new 2026-10-14)

U+2107

Dice



Dice —; niladic: single 6-sided die (1-6)

U+1D56F

Dice Percentile

‰

*Dice Percentile — niladic, intrinsic: A gets D10, B gets D10, yields + B * 10 A; the per mille sign U+2030 (2026-10-20)*

U+2030

D2



D2 — niladic: a single 2-sided die (1-2); sugar for 𝔇 2; the die face U+2681 (2026-10-20)

U+2681

D4



D4 — niladic: a single 4-sided die (1-4); sugar for 𝔇 4; the die face U+2683 (2026-10-20)

U+2683

D6



D6 — niladic: a single 6-sided die (1-6); sugar for 𝔇 6; the die face U+2685 (2026-10-20)

U+2685

D10



D10 — niladic: a single 10-sided die (1-10); sugar for 𝔇10; the circled ten U+2469 (2026-10-20)

U+2469

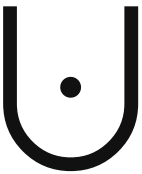
D12



D12 — niladic: a single 12-sided die (1-12); sugar for 𝔇12; the circled twelve U+246B (2026-10-20)

U+246B

Pick



Pick —; niladic: Nil

U+2A03

Constants and Chance: Monadic

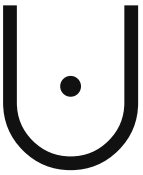
Half

½

Half —; monadic ½ X: X divided by 2; array → sugar for ⊞ (⍴ X); train → floor(length/2) index; dictionary → Nil (new 2026-10-13)

U+00BD

Pick



Pick —; monadic ⌣⊙X: scalar → same scalar; train → use Dice of size train to select element; array → use Dice of sides from shape to select element; dictionary → random {key:value} pair (new 2026-10-13)

U+2A03

Bell

𝔅

Bell —; monadic 𝔅 μ: bell curve point with mean μ (μ in (-1,1))

U+1D56D

Dice

𝔇

Dice —; monadic 𝔇 X: single X-sided die (1-X)

U+1D56F

Deal



Deal — given a scalar, works as expected; given an array, fills the entire back rank with limited values. Concrete examples and summary TBD (new 2026-10-11)

U+003F

Constants and Chance: Dyadic

Bell



Bell —; dyadic 𝔅 μ σ: bell curve point with mean μ and stddev σ (σ ≤ 1/3) (new 2026-10-13)

U+1D56D

Dice



Dice —; dyadic 𝔇 X N: N X-sided dice; all sugar via 𝔑 (hence 𝔅) (new 2026-10-13; die face U+2685 tried and reverted 2026-10-20)

U+1D56F

Number Theory: Monadic

Divisible



Divisible —; monadic ∣ X yields Truth when X is divisible by 2, Nil otherwise

U+23A2

Indivisible



Indivisible —; monadic ∤ X: sugar for ¬ ∣ X (the Odd test)

U+2AEE

Primes



Primes — monadic ℙ X yields the prime factors of X, per J's q:. (new 2026-10-18)

U+1D4AB

Prime



Prime — monadic 𝔓 N yields the Nth prime, per J's p: (zero-indexed: 𝔓 0 is 2). (new 2026-10-18)

U+1D513

Prime



Prime — monadic ⌽ X: sugar for ⍭⍘ X ⍳ 2 ⌊ √ X ⋮ ≤ X 1 (new 2026-10-15)

U+23C0

Factor



Factor — factorization verb (new 2026-10-16)

U+27D2

Combinatoric



Combinatoric —; monadic ## X: factorial of X

U+2A69

Number Theory: Dyadic

LCM



LCM — dyadic ⨶ X Y: least common multiple of X and Y (new 2026-10-16)

U+238A

Divisible



Divisible —; dyadic ∣ X Y yields Truth when X is divisible by Y, Nil otherwise (new 2026-10-15)

U+23A2

GCD



GCD — dyadic ⫻ X Y: greatest common divisor of X and Y (new 2026-10-16)

U+2980

Indivisible



Indivisible —; dyadic ∤ X Y: sugar for ¬ ∣ X Y (boolean Not of Divisible) (new 2026-10-15)

U+2AEE

Combinatoric



Combinatoric —; dyadic ⋕ X Y: binomial X choose Y (new 2026-10-16)

U+2A69

Polynomials and Roots: Monadic

NextLocal



NextLocal — the Roots division step: given P, yields [R prime, P prime] where R prime is Local(P) and P prime is P deflated by synthetic division of R prime. (new 2026-10-19)

U+2113

Roots



Roots — sugar: ℞ P is ℜ ℓ P (Reveal NextLocal over P); yields all polynomial roots. (new 2026-10-18; factored via Reveal 2026-10-19)

U+211E

Differentiate



Differentiate — train verb ∂ T yields symbolically differentiated version of train T of coefficients, multiplying constants as needed; same order as Polynomial (new 2026-10-13)

U+2202

Integrate



Integrate — train verb ∫ T yields symbolically integrated version of train T of coefficients, dividing constants as needed; same order as Polynomial (new 2026-10-13)

U+222B

Solve



Solve — monadic & A: array → vector of polynomial coefficients; missing input row is all zeroes (new 2026-10-13)

U+0026

Optimize



*Optimize —; monadic 𝔅 V: vector → polynomial; searches free variable with delta = 0.01 * domain, epsilon defaults to 1e-6*

U+214B

Local



Local — monadic ⍵ø V is 𝔅Ⅲ𝔏 & V (FixedPoint Optimize Triple Solve V); approximates best x for a local minimal of V (new 2026-10-13)

U+1D50F

Polynomials and Roots: Dyadic

Reveal



Reveal — intrinsic. Dyadic ℜ P Y yields the successive application of P upon Y as Y shrinks to Nil. Epsilon from P has two elements: an alpha to apply P to next Y, and the next Y. (new 2026-10-18)

U+211C

Optimize



Optimize —; dyadic ᗐ V ε: recurse into smaller domains until change in c < ε (new 2026-10-13)

U+214B

Integer Division



Integer Division — dyadic ⫽ X Y divides X by Y, yielding an integer (replaces Slash / integer division from manual p.19; / is now Reduce). Vector form: ⫽ V X where V is vector yields synthetic quotient of V by X (new 2026-10-14)

U+2AFD

Evaluate



Evaluate — dyadic 𝒇 X T evaluates train T of coefficients upon X (new 2026-10-13)

U+1D58B

FixedPoint



Fixed point — dyadic ꞇø V guess: refine guess until epsilon < 1e-6 (max 100 steps); failure to converge returns infinity (new 2026-10-13)

U+AB44

Matrices: Niladic

Outer Product



Outer Product —; Niladic ⊗ yields 1 (new 2026-10-12)

U+2297

Inner Product



Inner Product —; Niladic · yields 1 (new 2026-10-12; matrix case 2026-10-14)

U+22C5

Multiply Matrices



Multiply Matrices —; Niladic ⋈ yields the 1×1 identity matrix (new 2026-10-14)

U+22C8

Matrices: Monadic

Outer Product



Outer Product —; Monadic ⊗ X yields X.

U+2297

Inner Product



Inner Product —; Monadic · X yields X.

U+22C5

Multiply Matrices



Multiply Matrices —; Monadic ⋈ X yields X.

U+22C8

L Factor



L Factor — monadic: the L (lower triangular, 1s on diagonal) factor of the LU decomposition of a rank-2 array (new 2026-10-14)

U+1D577

Determinant



Determinant — monadic ⌹X: determinant of square matrix X (new 2026-10-16)

U+233C

U Factor



U Factor — monadic: the U (upper triangular) factor of the LU decomposition of a rank-2 array (new 2026-10-14)

U+1D580

Matrices: Dyadic

Outer Product



Outer Product —; dyadic ⊗ X Y yields the outer product matrix where entry [i,j] is × X[i] Y[j]; shape is the concatenation of the shapes of X and Y.

U+2297

Inner Product



Inner Product —; dyadic · X Y: for vectors X, Y, the dot product (sum of pairwise products); for matrices, the Frobenius inner product: a scalar, the sum of all pairwise products.

U+22C5

Multiply Matrices



Multiply Matrices —; dyadic ⋈ X Y: matrix multiplication, (⋈ X Y)[i;j] = · X[i;] (⍤ Y)[j;]; inner dimensions must agree, non-square fine.

U+22C8

Arrays: Niladic

Pee



Pee — Where⊡Pee yields Where [0, 1
(first two coordinates)

U+2119

Zee



Zee — the last position of a slice
(Where⊡Zee yields Where [-1)

U+2124

Exx



Exx — the zero position of a slice
(Where [0)

U+1D54F

Wye



Wye — the midposition of a slice

U+1D550

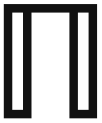
Extend



Extend —; niladic ⊢: sugar for ⊂ Nil

U+213E

Catenate



Catenate —; niladic ⊣: sugar for ⊂ Nil

U+213F

Grade Up



Grade Up — ⍋X yields the indices that sort X ascending. Niladic ⍋ yields 0 (from APL; new 2026-10-12)

U+234B

Grade Down



Grade Down — ⍒X yields the indices that sort X descending. Niladic ⍒ yields Nil (from APL; new 2026-10-12)

U+2352

Arrays: Monadic

Rank



Rank — monadic ⌗ X yields the number of dimensions in X; zero when scalar. (new 2026-10-18)

U+22D5

Permute



Permute —; monadic ⍨ X and dyadic X ⍨ Y work as in J's C.. Both modes intrinsic. (new 2026-10-19)

U+2A0A

Laminate



Laminate —; monadic ⍪ X: adds a leading singleton axis

U+2A5E

Anagram



Anagram — permutation index/parity (J's A.). Both modes intrinsic. (new 2026-10-19)

U+1D49C

Enumerate

№

Enumerate — monadic № X: sugar for ⍕⍳≢⍴X; yields index-value pairs for X (new 2026-10-18)

U+2116

Reflect



Reflect — monadic ÷X: sugar for ⍕⇋⍤-⇋⍤⍳⍤⍴X (new 2026-10-15)

U+223B

Count

77

⍳

U+2373

Kount



Kount — monadic ⍭B: takes a vector of T or Nil; returns a vector counting the T, turning each Nil to 0. Intrinsic. (new 2026-10-18)

U+2A23

⍳ Count

U+2373

Count — magical: when given a shape, fills an array matching shape. Scalar: behaves as normal. Array: fills array with next dimension limited by scalar from back rank — e.g. ⍳ 2 3 → rows of 0 1 2 (origin 0; last axis counts, outer axes replicate) (new 2026-10-11)

RavelRows



RavelRows — monadic ⍪ X: yields the rows of X as a vector (Antitable; new 2026-10-15)

U+22F2

Ravel



Ravel — monadic ⍎ X: flattens array X to vector, row-major (new 2026-10-14)

U+22F3

Reverse



Reverse —; monadic ⌽ X reverses elements along the last axis

U+233D

Transpose



Transpose — monadic ⍉ X reverses the order of axes (from APL; new 2026-10-12)

U+2349

Shape



Shape —; monadic ρ X: shape of X; scalar → 0; array → vector of dimensions; train → [1, length of train]; dictionary → [1, pairs present]

U+03C1

Extend



Extend —; monadic ⍐ X: Table X as in APL, adds trailing axis of length 1

U+213E

Catenate



Catenate —; monadic ⍪ X: catenates X to itself

U+213F

Rotate



Rotate —; monadic ⌽ X: rotate X toward origin by 1

U+29BD

Cardinality



Cardinality — "how big is this?"; scalar → -1; array → number of elements held; dictionary → number of pairs; train → length of train (new 2026-10-13)

U+2210

Center



Center — takes a shape, returns its integer halves expressed as a vector (e.g. [10, 6] → [5, 3]) (new 2026-10-13)

U+229E

Sight



Sight — takes 2D array, finds center, returns vector location of nearest non-zero scalar; ties pick one (new 2026-10-13)

U+2388

Arrays: Dyadic

Put



Put — ⫤⫥ X P A: value at coordinate P of array A becomes X (new 2026-10-18)

U+27DA

Permute



Permute —

U+2A0A

Laminate



Laminate —; dyadic ⩓ A B: yields [A; B]; when A and B are both scalars, yields A B (new 2026-10-17)

U+2A5E

Anagram



Anagram — permutation index/parity (J's A.). Both modes intrinsic. (new 2026-10-19)

U+1D49C

Catenate



Catenate —; dyadic ⫿ X Y: catenate X and Y as in APL; variadic ⫿ A ... Z: catenate A through Z left to right (new 2026-10-14)

U+213F

Take



Take — ↑ X Y takes the first X elements of Y (last X if X is negative); X may be a vector for per-axis counts (from APL; new 2026-10-12)

U+2191

Drop



Drop — ↓ X Y drops the first X elements of Y (last X if X is negative); X may be a vector for per-axis counts (from APL; new 2026-10-12)

U+2193

Reverse



Reverse —; dyadic ⌽ X Y rotates Y by X positions (from APL; new 2026-10-12)

U+233D

Shape

ρ

Shape —; dyadic ρ S X: reshape X to shape S (per APL) (new 2026-10-13)

U+03C1

Extend



Extend —; dyadic ⍓ X N: extends by N axes (new 2026-10-14)

U+213E

Rotate



Rotate —; dyadic ⌽̂ X Y: rotate X by Y (Y positive toward origin, negative away) (new 2026-10-16)

U+29BD

Boxes: Niladic

Box



Box —; Niladic ⊂ yields Truth
(from APL Enclose; new 2026-10-12)

U+2282

Unbox



Unbox —; Niladic ⊃ yields Nil
(from APL Disclose; new 2026-10-12)

U+2283

Summarize



Summarize — the summary effect at full power. Array case: find the 0,0 corner; if another array, recurse into the subarray; if a scalar, yield the boxed scalar. Scalar case yields the boxed scalar; niladic ⊚ is the 1,1 vector containing zero.

U+229A

Boxes: Monadic

Box



Box — monadic ⧆ A: wraps A in a box so it travels as a single element (new 2026-10-17)

U+2256

Unbox



Unbox — monadic ⧈ A: opens the box, yielding its contents (new 2026-10-17)

U+2257

Catalogue



Catalogue —; monadic ⊚X: cartesian product of the boxed lists in vector X (new 2026-10-17)

U+2A37

Raze



Raze — monadic ⫻A: sugar for Ravel Map Unbox A; opens every box and joins the contents into one vector (new 2026-10-17)

U+2AF5

Box



Box —; monadic ⊂ X encloses X into a scalar box.

U+2282

Unbox



Unbox —; monadic ⊃ X yields X.

U+2283

Summarize



Summarize — the summary effect at full power. Array case: find the 0,0 corner; if another array, recurse into the subarray; if a scalar, yield the boxed scalar. Scalar case yields the boxed scalar

U+229A

Zip

91



U+22B0



Zip

`U+22B0`

Zip — takes a train of vectors and zips them: ⍭ V1 V2 ... Vn yields an m×n matrix (m = vector length, n = train length); the resulting width is the length of the train; a scalar in the train expands into a column to fit the others; a container in the train contributes only its origin scalar, recursively (the summary effect); mismatched lengths Nil-pad (consistent with Diction); zips along the back rank for higher-rank arrays; ⍭ 1 2 yields the column [1;2] (new 2026-10-12; recycles the retired AntiSignum glyph)

Boxes: Dyadic

Summarize

94



U+229A

Fetch



Fetch — dyadic ⊸ V X: applies V as search vector into X's items; follows path V down through nested boxes (new 2026-10-17)

U+22B8

Catalogue



Catalogue —; dyadic ⊗ A B: yields [[A0,B0] ... [A1,B1]] ... [[AN,B0] ... [AN,BN]], the outer product matrix of pairs

U+2A37

Link



Link — dyadic ⫩ X Y: sugar for Catenate Map Box X Y; boxes X and Y into a two-element list (new 2026-10-17)

U+2AE9



Summarize

U+229A

Summarize — the summary effect at full power. Array case: find the 0,0 corner; if another array, recurse into the subarray; if a scalar, yield the boxed scalar. Scalar case yields the boxed scalar; Dyadic D ⊚ A: D (Direction) is a rank-1 array of coordinates; summarization starts at A[D], then proceeds as monadic; a scalar Direction s means start at (s, 0); Zip summarizes each train element (new 2026-10-12)

Box



Box —; Dyadic ⊂ X Y encloses X by Y levels; Y=0 leaves X unchanged.

U+2282

Unbox

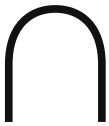


Unbox —; Dyadic ⊃ X Y picks element X from Y.

U+2283

Sets and Membership: Niladic

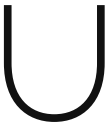
Intersection



Intersection —; niladic ∩: sugar for ⊂ Nil

U+2229

Union

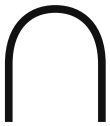


Union —; niladic ∪: sugar for ⊂ Nil

U+222A

Sets and Membership: Monadic

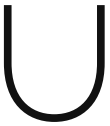
Intersection



Intersection —; monadic ∩ A: is A

U+2229

Union



Union —; monadic ∪ X: unique as in APL

U+222A

InOf



InOf — the curry of In: ⇔ InOf 𝐶 ∈ X; monadic predicate testing membership in X. (new 2026-10-18)

U+22F8

Where



Where — monadic ⍸Y: returns vector of indices of all elements of array Y that are Truth (new 2026-10-18)

U+2A42

Enlist

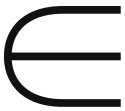


Enlist —; monadic ∈ Y: as defined in APL (deep flatten to vector)

U+220A

Sets and Membership: Dyadic

In



In — dyadic ∈ Y X yields Truth when X is in Y, Nil otherwise (J's e.). (new 2026-10-18)

U+2208

Bucket



Bucket — dyadic ∐ X Y: yields index of interval in X that Y falls into (new 2026-10-18)

U+2210

InLast



InLast — dyadic ∈̇ X Y: yields the index of the last occurrence of X in Y; yields length of Y if not found (new 2026-10-18)

U+22F5

InAt



InAt — dyadic ⋵ X Y: yields the index of Y whose value equals X; yields length of Y if not found (new 2026-10-18)

U+22F9

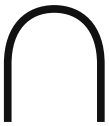
In



Enlist —; dyadic ∈ X Y: find/membership as defined in APL (new 2026-10-13)

U+220A

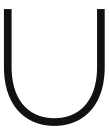
Intersection



Intersection —; dyadic ∩ A B: as in APL, elements in both (new 2026-10-14)

U+2229

Union



Union —; dyadic ∪ X Y: union as in APL (new 2026-10-14)

U+222A

Set Difference



Set Difference — dyadic ⍠A B: elements in A but not in B (new 2026-10-14)

U+2A43

Trains: Niladic

Reduce

/

Reduce — / F V: start at the highest index (its scalar is the initial result), step down toward origin; at each step result = F(scalar, result). Right fold.; Niladic / returns 0 (new 2026-10-12)

U+002F

Map



Map — ⟠ A B executes function A with zero arguments upon all scalars of B, as a back rank operator; usual back scalar rules.; Niladic ⟠ yields Truth, same as niladic 𝓘 (new 2026-10-12)

U+226C

Reduce (forward)

⌿

⌿ F V: start at the origin (its scalar is the initial result), step up toward the highest index; at each step result = F(result, scalar). Left fold, the reversed scan.

U+233F

Identity



Identity — 𝓘 V yields V unchanged; on a train, yields the train unchanged. Niladic 𝓘 yields Truth (new 2026-10-12)

U+1D574

Backstroke



Backstroke — Scan: the running Reduce, paired with /. \ F V on a length-n V yields a length-n vector where element i is / F (right fold) applied to the first i elements of V.; Niladic \ yields 0 (new 2026-10-12)

U+005C

Stop



Stop — statement separator; separates statements on one line. A carriage return is sugar for Stop (2026-10-20)

U+22C4

Backstroke (forward)



Backstroke (forward) — Reverse Scan, paired with ⌿. ⍀ F V on a length-n V yields a length-n vector where element i is ⌿F (left fold) applied to the first i elements of V.

U+2340

Strip



Strip —; niladic: removes the train from head up to but not including the first Stop encountered

U+2ACF

Comment



Comment — disregards the entire following train, up to and including Stop (2026-10-20)

U+270C

Trains: Monadic

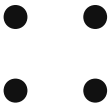
Meter

112



U+2214

Atomize



Atomize — monadic ∷ A: reads up to the first 32 elements of A (origin to end) and prepares them for use by train elements ahead (new 2026-10-14)

U+2237

Duplicate



Duplicate — copies a train to a train twice as long: ⊷ A B C → A B C A B C (was ==U+2A75, then ⁚ U+2236 — settled 2026-10-12)

U+22B7

Caboose



Caboose — picks up the train from head to Stop, then converts boxed contents to a vector (inverse of Atomize) (new 2026-10-15)

U+2A78



Meter

U+2214

Meter — measures its argument: ∔ V is ≺ + ⊷ V (zip the needle with the signal); yields M, a sequence of (needle, signal) pairs, needle ∈ -/0/+. M works in place of V in all Collate modes: the needle selects the function, the function transforms the signal (was ±, corrected to ≺ 2026-10-12)

Reduce



Reduce — / F V: start at the highest index (its scalar is the initial result), step down toward origin; at each step result = F(scalar, result). Right fold.; Monadic / V yields a linear array of the scalars ripped from V, highest to lowest indices.

U+002F

Map



Map — ⍁ A B executes function A with zero arguments upon all scalars of B, as a back rank operator; usual back scalar rules.; Monadic ⍁ B is identical to Identity ⍳ B.

U+226C

Reduce (forward)



⌿F V: start at the origin (its scalar is the initial result), step up toward the highest index; at each step result = F(result, scalar). Left fold, the reversed scan.; Monadic ⌿V and niladic ⌿ behave identically to / (new 2026-10-12)

U+233F

Zero



Zero — on a train, replaces all values with zeroes while keeping shapes (new 2026-10-12)

U+1D585

Backstroke

115



U+005C

Duplicate Head



Duplicate Head — ⥣ A ... yields A A ...; duplicates head of train, aka duplicate/dup (new 2026-10-14)

U+2161

Backstroke (forward)



Backstroke (forward) — Reverse Scan, paired with ⫻. ⟊ F V on a length-n V yields a length-n vector where element i is ⫻F (left fold) applied to the first i elements of V.; Monadic ⟊ V and niladic ⟊ behave identically to \ (new 2026-10-12)

U+2340

Pipeline



Pipeline —

U+2AF7



Backstroke

`U+005C`

Backstroke — Scan: the running Reduce, paired with /. \ F V on a length-n V yields a length-n vector where element i is / F (right fold) applied to the first i elements of V.; Monadic \ V yields a linear array of the scalars ripped from V, lowest to highest indices.

Head



Head — sugar for ⊐1 (new 2026-10-15)

U+2991

Tail



Tail — sugar for ⊐-2 (new 2026-10-15)

U+2992

Duplicate Tail



Duplicate Tail — ⧟... A yields ... A A; duplicates tail of train (new 2026-10-13)

U+29DE

Strip

117



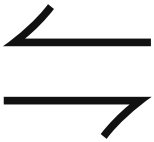
U+2ACF

⊐ Strip

U+2ACF

Strip —; monadic ⊐ X: removes the train from position X (0 is head) up to but not including the first Stop encountered; when X < 0, strips from position X in reverse order back to head, where −1 is the final tail position adjacent to the first Stop encountered (new 2026-10-15)

Swap



Swap — ⇋ A B ... yields B A ...; swaps head two of train. Single: ⇋ A Stop X yields X. Bare: ⇋ Stop X yields X (new 2026-10-13)

U+21CB

Replicate



U+232C

Rotate Head



Rotate Head — ⟲ A B C ... yields B C ... A; rotates head to tail (new 2026-10-13)

U+27F2

Rotate Tail



Rotate Tail — ⟳ ... A B C yields C ... A B; rotates tail to head (new 2026-10-13)

U+27F3



Replicate

U+232C

Replicate — ⌬ N X Stop yields X replicated N times; e.g. ⌬ 5 A Stop gives A A A A A. Zero case: ⌬ 0 A Stop X yields X. Negative N is antireplicate: ⌬ -1 A A A A A Stop yields A A A A. Empty antireplicate: ⌬ -1 Stop X yields X. Bare: ⌬ Stop X yields X (new 2026-10-13)

Nip



Nip — ↤ A B ... yields A ...; drops second, keeps head (new 2026-10-13)

U+21F7

Tuck



Tuck — ↦ A B ... yields A B A ...; copies head under second (new 2026-10-13)

U+21F8

Reverse Train



Reverse Train — reverses order of train (new 2026-10-13)

U+21F9

Over



Over — ⤺ A B ... yields B A B ...; copies second to head (new 2026-10-13)

U+293A

Triple

Ⅲ

Triple — Ⅲ X yields X X X; triplicates top of stack (new 2026-10-13)

U+2162

Fork



Fork — ⋔ F G H X: sugar for G H Swap F Dup X (J fork; new 2026-10-14)

U+22D4

Drop Train



Drop Train — sugar for reverse, replicate -1, reverse; drops head of train (new 2026-10-13)

U+291A

Trains: Dyadic

Applyself



Applyself — ⍣⍤1 f n a applies f n times to a; sugar for ⫷a ⍵⊂⍣⍺⍵ f n (from APL power; new 2026-10-12)

U+2363

Pipeline



Pipeline —; dyadic ⫷X Y applies vector Y, readable as a train of monadic functions, in sequence to X (new 2026-10-12)

U+2AF7

Functions: Monadic

Diction



Diction — dict-backed decision-table constructor (was Vectorize, renamed 2026-10-11). ¡ A B → {A:B}; ¡ A B C D → {A:B,C:D}; monadic ¡ (a b c) self-maps into a set. Membership sugar: X member Y → ⊤ (X ¿ (¡ Y))

U+00A1

Glyph



Glyph — the name for a single character value. Glyph ← ≡ ⌧: Glyph's function is the literalize train (via monadic ≡) — to make a Glyph, literalize

U+00AE

Decide

127



U+00BF

Function



Function — gets glyph del (∇, nabla): the function-definition operator. Monadic ∇ Y executes array Y as an FSA. (monadic 2026-10-19)

U+2207

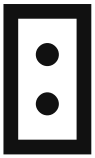


Decide

U+00BF

Decide — first-match generalized switch/match over n×2k train tables (was When, 2026-10-11). n×2 returns 1×1 (scalar); n×4 returns 1×2; generally n×2k returns 1×k. An explicit trailing C is unmatched-result filler, not an else branch; without filler, unmatched returns Nil without logging an error. Canonical: R ← X ¿ ¡ A B C

Print



Print — the quad-colon verb: prints the value from its right, then passes that value to its left (new 2026-10-11)

U+2360

(literalize)



Literalize next nonspace char, or ※0x???? by codepoint

U+2A33

Rail



Rail — monadic ⌐X yields a vector of symbols from train X (train-to-array conversion). (new 2026-10-19)

U+2ACD

Lay



Lay — monadic ¬V yields the train laid down from origin to limit of vector V (array-to-train, inverse of Rail). (new 2026-10-19)

U+2ACE

Lambda

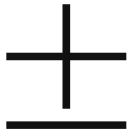


Lambda — lambda abstraction (new 2026-10-13)

U+039B

Functions: Dyadic

Collate



Collate — executes one of three trains upon a value: for less than zero, for zero, or for greater than zero (was AntiSignum ⫷ U+22B0, renamed 2026-10-12; complement to signum)

U+00B1

Curry

132



U+1D49E

$\mathcal{C}$ Curry

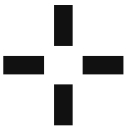
U+1D49E

Curry — intrinsic. Partial application: turns a dyadic verb into a monadic predicate by binding an argument, e.g. making = X from =. Semantics: Gets Q Curry P args: args to omega prime, omega prime shifts into alpha prime (binding the leftmost argument, textbook curry order), Q X makes X alpha, gets slice -1 of omega prime A, then call P; Q alpha yields P evaluated upon alpha prime, omega prime, and alpha as arguments in that order. Omega becomes nil upon Q's execution. Principle: Gets Q Curry P train is the principle for creating curries (Q is an example name assigned to the curried function, not a verb). (new 2026-10-18)

Errors: Niladic

Silence

136



U+22B9

Compress

137



U+22D6

Errline



Last error: which line

U+274C

Errmsg



Last error: what (type)

U+274E



Silence

U+22B9

Silence — ⊹ Z V drops all nonzero values; Z transforms the zero contents in sequence. ⊹ Z M on metered M: a nonzero needle drops the pair; a zero needle selects Z (applied to the signal).; Niladic ⊹ yields boxed Nil, a trial vector for enclosing operators to work on for starts (the dual of Compress; new 2026-10-12)



Compress

U+22D6

Compress — ⋖ P V applies transform P to each nonzero content of V; zero contents are eliminated; sequence order preserved. ⋖ P M on metered M: a nonzero needle selects P (applied to the signal); a zero needle drops the pair.; Niladic ⋖ yields boxed Truth, a trial vector for enclosing operators to work on for starts (a comprehension filter; new 2026-10-12)

Cut

Cut —; niladic: on error, erases the error and returns Nil

U+22EE

Debugger

✔⋄ *enters debugger iff error registers set*

U+2714

Errdmd

Last error: diamond position in line

U+2727

Succeed

Succeed — like Cut, except returns Truth (new 2026-10-15)

U+2999

Anticut



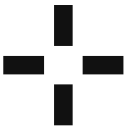
Anticut — on error throws to debugger; on return from debugger, returns Nil

U+22EF

Errors: Monadic

Silence

142



U+22B9

Compress

143



U+22D6

Cut



Cut —; monadic ⋮ X: checks X for Truth before returning

U+22EE



Silence

U+22B9

Silence — ⊹ Z V drops all nonzero values; Z transforms the zero contents in sequence. ⊹ Z M on metered M: a nonzero needle drops the pair; a zero needle selects Z (applied to the signal).; Monadic ⊹ V or ⊹ M means Z is identity: keep zeros (or zero-needle signals) as-is, drop nonzeros.



Compress

`U+22D6`

Compress — ⋖ P V applies transform P to each nonzero content of V; zero contents are eliminated; sequence order preserved. ⋖ P M on metered M: a nonzero needle selects P (applied to the signal); a zero needle drops the pair.; Monadic ⋖ V or ⋖ M means P is identity: pure filter, no transform.

Classes and Properties: Niladic

Constant

147

α

U+03B1

Epsilon

ε

Epsilon — holds the result of any getter, setter, or nabla when no explicit return glyph is used

U+03B5

Omega

ω

Omega — the vector of arguments passed to a setter (◅)

U+03C9

Person



Person — a callable class that emulates a person, like Muse, on the other end of a pipe: the factoring point for callbacks based in the person's responses (2026-10-20)

U+4EBA

α Constant

U+03B1

Constant — declares a class constant: X α name ← value names a constant for class X. α is also the original (pre-override) value: read-only, may not be changed directly; trying registers an error. As the original value α is an ad hoc constant — contextual to the setter, not a declared class constant

Classes and Properties: Monadic

Increment



Increment —; monadic ⊕ D: invokes D's setter with nabla + 1 as the xthrement.

U+2295

Setter



Setter — the setter nabla: defines a property setter, as ∇ defines a function. Generalized: takes an xthrement argument — modifies instance data (increment/decrement generalized) rather than assigning (new 2026-10-14)

U+22B2

Getter



Getter — the getter nabla: defines a property getter, as ∇ defines a function

U+22B3

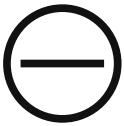
Property



Property —; Monadic ⋘ D: yields the result of D's getter (getter mode of D's class call) (new 2026-10-14)

U+22D8

Decrement



Decrement —; monadic ⊖ D: invokes D's setter with nabla - 1 as the xthrement.

U+2296

Divide By



Divide By —; monadic ⊘ D: invokes D's setter with nabla ÷ 2 as the xthrement (halve).

U+2298

Multiply By



Multiply By —; monadic ⊛ D: invokes D's setter with nabla × 2 as the xthrement (double).

U+229B

Classes and Properties: Dyadic

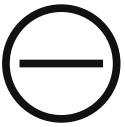
Increment



Increment —; Dyadic ⊕ D X: invokes D's setter with nabla + X as the xthrement (new 2026-10-14)

U+2295

Decrement



Decrement —; Dyadic ⊖ D X: invokes D's setter with nabla - X as the xthrement (new 2026-10-14)

U+2296

Multiply By



Multiply By —; Dyadic ⊛ D X: invokes D's setter with nabla × X as the xthrement (new 2026-10-14)

U+229B

Property



Property —; dyadic ⋘ I D: invokes D's getter (getter mode of D's class call), then invokes I's setter (setter mode of I's class call) with the result as argument; yields the assigned data as a pass-through.

U+22D8

Divide By



Divide By —; Dyadic ⊘ D X: invokes D's setter with nabla ÷ X as the xthrement (new 2026-10-14)

U+2298

(method binding)



Binds method to object: X⊡function

U+22A1

Gets



Gets — ⇔ A G: A stored in symbols[G], G stored in instances[A]; bidirectional binding (new 2026-10-13)

U+2B04

The Virtual: Niladic

IntegerBounded



IntegerBounded — bounded integer class: min/max enforced on creation, assignment, math; clamp + silent sticky flag on violation (new 2026-10-15)

U+012A

Trapdoor



Trapdoor — the assembly trap door: invokes machine code at a given address (installed via ctypes; proven 2026-10-14).

U+1D21D

VirtualSpace



VirtualSpace — the VR coordinate space: navel origin, forward half-sphere, [r;θ;φ], angles ±32767, distance in mm (new 2026-10-15)

U+1D4E5

Prim



Prim — base VR primitive: name, position, collision machinery; posted via Virtual (new 2026-10-16)

U+1D7D8

Raster



Raster — VR prim ⊑ Prim: array of pels, rank 4 [Y,X,1,3] (new 2026-10-16)

U+1D7D9

Point



Point — VR prim ⊑ Prim: dot prim, [N,6] (H,V,D,RGB) (new 2026-10-16)

U+1D7DA

Triangle



Triangle — VR prim ⊑ Prim: 9 points × 2 thetas + D + RGB = 22 numbers, [N,22] (new 2026-10-16)

U+1D7DB

The Virtual: Monadic

Trapdoor

160



U+1D21D

⊂ Trapdoor

`U+1D21D`

Trapdoor — the assembly trap door: invokes machine code at a given address (installed via ctypes; proven 2026-10-14).; Monadic ⊂ A: puts the address of A into the accumulator before calling. ⇔ X ⊂ ∇ assembly ∇: X gets the address of the compiled assembly script. If X is so bound, niladic X calls the code. ε maps to the accumulator. The call yields ε

Dungeon

Dungeon: Places

Where

@

Where the hero is — bound to '@', the roguelike player glyph. Where⊡Zee yields Where [-1 (last); Where⊡Wye yields Where [1 (midposition — middle of 3 coords); Where⊡Exx yields Where [0 (zero). INDICES START AT ZERO

U+0040

Dungeon

£

The Dungeon — literally the set of all levels

U+00A3

Screen

§

The Screen — intrinsic 20×60 array (16 play + 4 I/O rows)

U+00A7

Level

¶

Level — the pilcrow: a single dungeon level, member of £ Dungeon

U+00B6

Cell

¢

A dungeon cell — the cent sign joins the family: § £ ¢. Cell⊡location ← £ Where: the cell's location is the Dungeon at Where. A dungeon square IS a dictionary of instances (the ⓘ registry's entries present at that square)

U+00A2

Up



Up — the upward/north movement. Where⊡up yields ∅ when Where⊡Zee is 0, else Where with Zee decremented (refined 2026-10-10: boundary-checked — top edge is a quiet Nil, not a negative index)

U+25B2

Down



Down — the downward/south movement, counterpart to Up. Where⊡down yields ∅ when Where⊡Zee ≥ Dungeon⊡Zee (bottom edge), else Where⊡Zee + 1 (confirmed 2026-10-10)

U+25BC

Dungeon: Classes

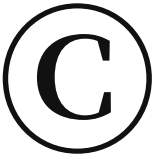
Class



Class — the circled dot. X⊡Class gets a RANK-2 ARRAY OF TRAINS: a class IS a decision table made of trains. Class instances are callable; the train defaults to Nil (∅)

U+2299

Classes



Classes — the set of all classes (plural of Class), parallel to £ Dungeon as the set of all levels. Implementation: a single twin (bidirectional word↔glyph) dictionary at global scope

U+24B8

Instances

168



U+24BE

Instance



Instance — the INDIVIDUAL instance (singular), distinct from Ⓘ the table. Joins the blackboard family (𝕏 𝕐 ℤ ℙ 𝕆). Carries the (glyph, hash-at-creation) identity tuple

U+1D540



Instances

U+24BE

Instances — the TABLE/set of all instances (plural), parallel to Ⓒ Classes. Implementation: a single twin (bidirectional word ↔ glyph) dictionary at global scope. Instance identity is the (glyph, hash-at-creation) tuple, used as the key field in both the global instance dictionary and each square (Cell) dictionary

Interpreter



Interpreter — the Wuxia interpreter (new 2026-10-15)

U+203D

Inherits



Inherits — the inheritance operator (square subset of)

U+2291

Isa



Isa — the is-a / instantiation operator (2026-10-10). Monadic ∈ Y: Y is a class to copy data from; new instance.ε = Y.ε. Dyadic ∈ Y X: X is a class, optional initializer; new instance.ε = X.ε (new 2026-10-14)

U+22FF

Object



Object — a DUNGEON object, not a generic OOP object. The blackboard O joins the projection family (𝕏 𝕐 ℤ ℙ): the root of the dungeon's object hierarchy, what every dungeon entity inherits from

U+1D546

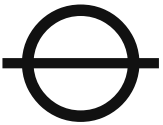
VirtualDistance



VirtualDistance — distance in VirtualSpace, 0-512 (new 2026-10-15)

U+27F0

Angle



Angle — a VR angle (theta/phi) (new 2026-10-15)

U+29B5

VirtualHorizontal



VirtualHorizontal — horizontal angle in VirtualSpace (new 2026-10-15)

U+29D3

VirtualVertical



VirtualVertical — vertical angle in VirtualSpace (new 2026-10-15)

U+29D7

Dungeon: Objects

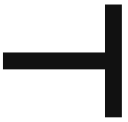
Ring



Ring — the blackboard R: a dungeon object (ring), third of the dungeon entity glyphs

U+211D

Wand



Wand — left tack: a dungeon object (wand)

U+22A3

Armor



Armor — the blackboard A: a dungeon object (armor), first of the dungeon entity glyphs

U+1D538

Weapon



Weapon — the blackboard W: a dungeon object (weapon), second of the dungeon entity glyphs

U+1D54E

Amulet



Amulet — eight-petalled floret: a dungeon object (amulet). (Was U+1DF76, tofu on device — swapped 2026-10-10)

U+2741

Food



Food — the blackboard F: a dungeon object (food)

U+1D53D

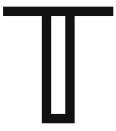
Scroll



Scroll — the blackboard S: a dungeon object (scroll)

U+1D54A

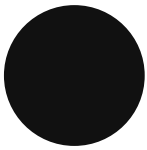
Tool



Tool — the blackboard T: a dungeon object (tool)

U+1D54B

Gem



Gem — solid black circle: a dungeon object (gem). (Was U+2733 — swapped 2026-10-10)

U+25CF

Potion



Potion — heavy exclamation: a dungeon object (potion)

U+2757

Dungeon: Entities

Player



Player — black smiling face: the dungeon's player entity. Inherits Monster plus a few extra fields Monster does not have

U+263B

Monster



Monster — the blackboard M: a dungeon monster, first of the monster glyphs

U+1D544

Dungeon: Food

Tin



Tin — circled white star: a dungeon food item (tin), first of the object species

U+272A

RefriedBeans



RefriedBeans — dark-encircled six: grants teleportation

U+277B

Spinach



Spinach — dark-encircled seven: Popeye strength

U+277C

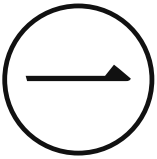
Sushi



Sushi — circled congratulation: the sushi/sashimi food category. (Was U+2722 — swapped 2026-10-11)

U+3297

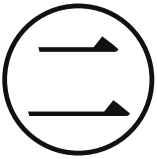
CarpSashimi



CarpSashimi — circled kanji: sushi item

U+3280

WhaleSashimi



WhaleSashimi — circled kanji: sushi item

U+3281

EelSashimi



EelSashimi — circled kanji: sushi item

U+3282

SalmonSashimi



SalmonSashimi — circled kanji: sushi item

U+3283

TunaSashimi



TunaSashimi — circled kanji: sushi item

U+3284

CrabSashimi



CrabSashimi — circled kanji: sushi item

U+3285

DragonSashimi



DragonSashimi — circled kanji: sushi item

U+3286

TurtleSashimi



TurtleSashimi — circled kanji: sushi item

U+3287

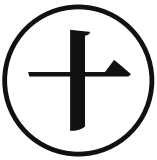
Oyster



Oyster — circled kanji: sushi item

U+3288

Uni



Uni — circled kanji: sushi item

U+3289

Gyuniku



Gyuniku — circled kanji: sushi item

U+328A

Asian



Asian — squared hand: the Asian food category. (Was U+328C — swapped 2026-10-11)

U+1F210

Ramen



Ramen — squared cjk unified ideograph-5b57: Asian food

U+1F211

FortuneCookie



FortuneCookie — squared cjk unified ideograph-53cc: Asian food

U+1F212

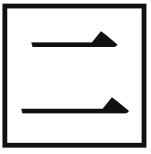
Kelp



Kelp — squared katakana de: Asian food

U+1F213

Dumplings



Dumplings — squared cjk unified ideograph-4e8c: Asian food

U+1F214

BowlOfRice



BowlOfRice — squared cjk unified ideograph-591a: Asian food

U+1F215

SpringRolls



SpringRolls — squared cjk unified ideograph-89e3: Asian food

U+1F216

Tofu



Tofu — squared cjk unified ideograph-5929: Asian food

U+1F217

PreservedEgg



PreservedEgg — squared cjk unified ideograph-4ea4: Asian food

U+1F218

Soup



Soup — circled twenty: the soup category

U+2473

Deli



Deli — black diamond minus white X: the delicatessen food category (the 25 staples, names lost 2026-10-10 — rebuild pending)

U+2756

Fruit



Fruit — domino tile back: the fruit food category

U+1F030

CantoneseDuck



CantoneseDuck — squared cjk unified ideograph-6620: Asian food

U+1F219

Egg

ø

Egg — o with stroke: the egg. (Was U+2800 — swapped 2026-10-11)

U+00F8

Meat



Meat — CJK beautiful: the meat category

U+4E3D

Creature



Creature — o with long stroke overlay: the monster-shuffle category

U+A74B

Candy



Candy — mahjong joker: the candy category

U+1F02A

Goodbar



Goodbar — mahjong east wind: candy

`U+1F000`

Honee



Honee — mahjong south wind: candy

`U+1F001`

DRation



DRation — mahjong west wind: candy

`U+1F002`

DoveBar



DoveBar — mahjong north wind: candy

`U+1F003`

AcidBroth

❶

AcidBroth — dark-encircled number: soup

U+2776

GoldenSoup

❷

GoldenSoup — dark-encircled number: soup

U+2777

StonedSoup

❸

StonedSoup — dark-encircled number: soup

U+2778

CrowBar



CrowBar — mahjong winter: candy

U+1F029

MoldySoup



MoldySoup — dark-encircled number: soup

U+2779

Manwich



Manwich — dark-encircled number: soup

U+277A

CreamOfAlgaeSoup



CreamOfAlgaeSoup — dark-encircled number: soup

U+277E

ChickenNoodleSoup



ChickenNoodleSoup — dark-encircled number: soup

U+277F

Apple



Apple — domino tile: fruit, inherits Fruit

U+1F032

Orange



Orange — domino tile: fruit, inherits Fruit

U+1F033

Pear



Pear — domino tile: fruit, inherits Fruit

U+1F034

Melon



Melon — domino tile: fruit, inherits Fruit

U+1F035

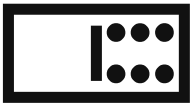
Banana



Banana — domino tile: fruit, inherits Fruit

U+1F036

Carrot



Carrot — domino tile: fruit, inherits Fruit

U+1F037

Fig



Fig — domino tile: fruit, inherits Fruit

U+1F038

Lemon



Lemon — domino tile: fruit, inherits Fruit

U+1F039

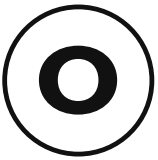
Tomb



Tomb — circled 10 on black square: safe-row tin, now Sundry

U+3248

Sundry



Sundry — circled ideograph: the sundry/miscellaneous category

U+3267

Lime



Lime — domino tile: fruit, inherits Fruit

U+1F03A

Grapes



Grapes — domino tile: fruit, inherits Fruit

U+1F03B

Pesto



Pesto — circled 20 on black square: safe-row tin, now Sundry

U+3249

KelpFrond



KelpFrond — circled water: sundry item (kelp is seaweed)

U+328C

EucalyptusLeaf



EucalyptusLeaf — circled wood: sundry item

U+328D

CandyBar



CandyBar — circled metal: sundry item

U+328E

LembasWafer



LembasWafer — circled earth: sundry item

U+328F

CramRation



CramRation — circled sun: sundry item

U+3290

FoodRation



FoodRation — circled medicine: sundry item

U+32A9

KRation



KRation — circled religion: sundry item

U+32AA

CloveOfGarlic



CloveOfGarlic — circled attention: sundry item

U+329F

SprigOfWolfsbane



SprigOfWolfsbane — circled item: sundry item

U+32A0

LumpOfRoyalJelly



LumpOfRoyalJelly — circled copy: sundry item

U+32A2

CRation



CRation — circled study: sundry item

U+32AB

Dungeon: Species inheritance

EarthElemental



⊑ *Elementals — Species inherits its category (2026-10-10)*

U+2295

FireElemental



⊑ *Elementals — Species inherits its category (2026-10-10)*

U+229B

AirElemental



⊑ *Elementals — Species inherits its category (2026-10-10)*

U+2A37

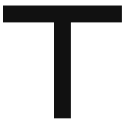
WaterElemental



⊑ *Elementals — Species inherits its category (2026-10-10)*

U+2A38

PurpleWorm



⊑ *Worms — Species inherits its category (2026-10-10)*

U+22A4

BabyPurpleWorm



⊑ *Worms — Species inherits its category (2026-10-10)*

U+22A6

LongWorm



⊑ *Worms — Species inherits its category (2026-10-10)*

U+2A53

BabyLongWorm



⊑ *Worms — Species inherits its category (2026-10-10)*

U+2A55

LargeKobold



⊑ *Kobolds — Species inherits its category (2026-10-10)*

U+049C

Kobold



⊑ *Kobolds — Species inherits its category (2026-10-10)*

U+049D

MasterMindFlayer



⊑ *Mindflayers — Species inherits its category (2026-10-10)*

U+04F4

MindFlayer



⊑ *Mindflayers — Species inherits its category (2026-10-10)*

U+04F5

StoneGiant

Ġ

⊑ *Giants — Species inherits its category (2026-10-10)*

U+0120

Giant

Ģ

⊑ *Giants — Species inherits its category (2026-10-10)*

U+0122

KoboldLord

Ќ

⊑ *Kobolds — Species inherits its category (2026-10-10)*

U+040C

KoboldShaman

Ҝ

⊑ *Kobolds — Species inherits its category (2026-10-10)*

U+049B

FireGiant

Ĝ

⊑ *Giants — Species inherits its category (2026-10-10)*

U+011C

HillGiant

Ğ

⊑ *Giants — Species inherits its category (2026-10-10)*

U+011E

FrostGiant

Ǵ

⊑ *Giants — Species inherits its category (2026-10-10)*

U+0193

Ettin

Gᵛ

⊑ *Giants — Species inherits its category (2026-10-10)*

U+01E6

VampireBat



⊑ *Vampires — Species inherits its category (2026-10-10)*

`U+0180`

StormGiant



⊑ *Giants — Species inherits its category (2026-10-10)*

`U+01E4`

Vampire



⊑ *Vampires — Species inherits its category (2026-10-10)*

`U+1E7F`

Titan



⊑ *Giants — Species inherits its category (2026-10-10)*

`U+2141`

VampireLeader



⊑ *Vampires — Species inherits its category (2026-10-10)*

U+1E7D

VampireLady



⊑ *Vampires — Species inherits its category (2026-10-10)*

U+2047

VampireLord



⊑ *Vampires — Species inherits its category (2026-10-10)*

U+2047

Nosferatu



⊑ *Vampires — Species inherits its category (2026-10-10)*

U+2047

HumanMummy



⊑ *Undead — Species inherits its category (2026-10-10)*

U+046A

ElfMummy



⊑ *Undead — Species inherits its category (2026-10-10)*

U+046A

DwarfMummy



⊑ *Undead — Species inherits its category (2026-10-10)*

U+046A

GnomeMummy



⊑ *Undead — Species inherits its category (2026-10-10)*

U+046A

OrcMummy



⊑ *Undead — Species inherits its category (2026-10-10)*

U+046A

KoboldMummy



⊑ *Undead — Species inherits its category (2026-10-10)*

U+046A

GiantMummy



⊑ *Undead — Species inherits its category (2026-10-10)*

U+046A

EttinMummy



⊑ *Undead — Species inherits its category (2026-10-10)*

U+046A

KoboldZombie



⊑ *Undead — Species inherits its category (2026-10-10)*

U+2124

GnomeZombie



⊑ *Undead — Species inherits its category (2026-10-10)*

U+2124

OrcZombie



⊑ *Undead — Species inherits its category (2026-10-10)*

U+2124

DwarfZombie



⊑ *Undead — Species inherits its category (2026-10-10)*

U+2124

ElfZombie



⊑ *Undead — Species inherits its category (2026-10-10)*

U+2124

HumanZombie



⊑ *Undead — Species inherits its category (2026-10-10)*

U+2124

GiantZombie



⊑ *Undead — Species inherits its category (2026-10-10)*

U+2124

EttinZombie



⊑ *Undead — Species inherits its category (2026-10-10)*

U+2124

Ghoul

g

⊑ *Undead — Species inherits its category (2026-10-10)*

U+210A

WoodElf



⊑ *Elves — Species inherits its category (2026-10-10)*

U+2618

GreenElf



⊑ *Elves — Species inherits its category (2026-10-10)*

U+2618

GreyElf



⊑ *Elves — Species inherits its category (2026-10-10)*

U+2618

BabyDragon

⁇

⊑ *Dragons — Species inherits its category (2026-10-10)*

U+2047

Elf



⊑ *Elves — Species inherits its category (2026-10-10)*

U+2618

ElfNoble



⊑ *Elves — Species inherits its category (2026-10-10)*

U+2E19

ElvenMonarch

支

⊑ *Elves — Species inherits its category (2026-10-10)*

U+2F40

Centipede

¢

⊑ *Spiders — Species inherits its category (2026-10-10)*

U+00A2

CaveSpider

¤

⊑ *Spiders — Species inherits its category (2026-10-10)*

U+00A4

Scorpion

±

⊑ *Spiders — Species inherits its category (2026-10-10)*

U+00B1

Dragon

⁇

⊑ *Dragons — Species inherits its category (2026-10-10)*

U+2047

GiantSpider

Ø

⊑ *Spiders — Species inherits its category (2026-10-10)*

U+00D8

Mastodon

Ψ

⊑ *Beasts — Species inherits its category (2026-10-10)*

U+03A8

Mumak

Ψ

⊑ *Beasts — Species inherits its category (2026-10-10)*

U+03A8

Baluchitherium

ϒ

⊑ *Beasts — Species inherits its category (2026-10-10)*

U+03D2

Wumpus

¿

⊑ *Beasts — Species inherits its category (2026-10-10)*

U+00BF

Trapper

Й

⊑ *Beasts — Species inherits its category (2026-10-10)*

U+048A

Minotaur

Ṁ

⊑ *Beasts — Species inherits its category (2026-10-10)*

U+1E40

Titanothere

ơ

⊑ *Beasts — Species inherits its category (2026-10-10)*

U+213A

Bugbear

ö

⊑ *Beasts — Species inherits its category (2026-10-10)*

U+00F6

Owlbear

ø

⊑ *Beasts — Species inherits its category (2026-10-10)*

U+00F8

Jabberwock

ƹ

⊑ *Beasts — Species inherits its category (2026-10-10)*

U+01B8

Rothe

ȣ

⊑ *Beasts — Species inherits its category (2026-10-10)*

U+0222

Shade

@

⊑ *Guises — Species inherits its category (2026-10-10)*

U+0040

Leocrotta

ꝉ

⊑ *Beasts — Species inherits its category (2026-10-10)*

U+023D

Stalker

Ǝ

⊑ *Lurkers — Species inherits its category (2026-10-10)*

U+042D

LurkerAbove



⊑ *Lurkers — Species inherits its category (2026-10-10)*

U+214F

Skeleton

@

⊑ *Guises — Species inherits its category (2026-10-10)*

U+0040

Foocubus

@

⊑ *Guises — Species inherits its category (2026-10-10)*

U+0040

MountainNymph

@

⊑ *Guises — Species inherits its category (2026-10-10)*

U+0040

WoodNymph

@

⊑ *Guises — Species inherits its category (2026-10-10)*

U+0040

WaterNymph



⊑ *Guises — Species inherits its category (2026-10-10)*

U+0040

MerfolkRoyal



⊑ *Merfolk — Species inherits its category (2026-10-10)*

U+2047

MerfolkNoble



⊑ *Merfolk — Species inherits its category (2026-10-10)*

U+2047

MerfolkWitch



⊑ *Merfolk — Species inherits its category (2026-10-10)*

U+2047

WizardOfYendor

¥

⊑ *Specials — Species inherits its category (2026-10-10)*

U+00A5

Shopkeeper

§

⊑ *Specials — Species inherits its category (2026-10-10)*

U+00A7

MerfolkCommon

⁇

⊑ *Merfolk — Species inherits its category (2026-10-10)*

U+2047

GrandMaster

⁇

⊑ *Specials — Species inherits its category (2026-10-10)*

U+2047

Tin



⊑ *Food — Tin inherits Food — first object species joins the hierarchy (2026-10-10)*

U+272A

AcidBroth



⊑ *Soup — Soup inherits Soup (2026-10-10)*

U+2776

GoldenSoup



⊑ *Soup — Soup inherits Soup (2026-10-10)*

U+2777

StonedSoup



⊑ *Soup — Soup inherits Soup (2026-10-10)*

U+2778

MoldySoup

4

⊑ *Soup — Soup inherits Soup (2026-10-10)*

U+2779

Manwich

5

⊑ *Soup — Soup inherits Soup — the human-flesh tin (2026-10-10)*

U+277A

CreamOfAlgaeSoup

9

⊑ *Soup — Soup inherits Soup (2026-10-10)*

U+277E

ChickenNoodleSoup

10

⊑ *Soup — Soup inherits Soup (2026-10-10)*

U+277F

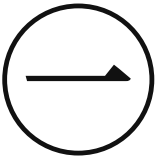
RefriedBeans



⊑ *Soup — Soup item inherits Soup — grants teleportation (2026-10-11)*

U+277B

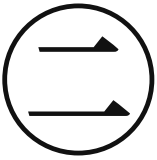
CarpSashimi



⊑ *Sushi — Individual sushi inherits Sushi (2026-10-10)*

U+3280

WhaleSashimi



⊑ *Sushi — Individual sushi inherits Sushi (2026-10-10)*

U+3281

Sushi



⊑ *Food — Sushi inherits Food — the sushi food category (2026-10-10)*

U+3297

EelSashimi



⊑ *Sushi — Individual sushi*
inherits Sushi (2026-10-10)

U+3282

SalmonSashimi



⊑ *Sushi — Individual sushi*
inherits Sushi (2026-10-10)

U+3283

TunaSashimi



⊑ *Sushi — Individual sushi*
inherits Sushi (2026-10-10)

U+3284

CrabSashimi



⊑ *Sushi — Individual sushi*
inherits Sushi (2026-10-10)

U+3285

DragonSashimi



⊑ *Sushi — Individual sushi inherits Sushi (2026-10-10)*

U+3286

TurtleSashimi



⊑ *Sushi — Individual sushi inherits Sushi (2026-10-10)*

U+3287

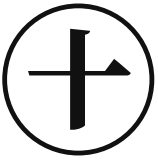
Oyster



⊑ *Sushi — Individual sushi inherits Sushi (2026-10-10)*

U+3288

Uni



⊑ *Sushi — Individual sushi inherits Sushi (2026-10-10)*

U+3289

Gyuniku



⊑ *Sushi — Individual sushi inherits Sushi (2026-10-10)*

U+328A

Asian



⊑ *Food — Asian inherits Food — the Asian food category (2026-10-10)*

U+1F210

Ramen



⊑ *Asian — Individual Asian food inherits Asian (2026-10-10)*

U+1F211

FortuneCookie



⊑ *Asian — Individual Asian food inherits Asian (2026-10-10)*

U+1F212

Kelp



⊑ *Asian — Individual Asian food inherits Asian (2026-10-10)*

U+1F213

Dumplings



⊑ *Asian — Individual Asian food inherits Asian (2026-10-10)*

U+1F214

BowlOfRice



⊑ *Asian — Individual Asian food inherits Asian (2026-10-10)*

U+1F215

SpringRolls



⊑ *Asian — Individual Asian food inherits Asian (2026-10-10)*

U+1F216

Deli



⊑ *Food — Deli inherits Food — the delicatessen food category (2026-10-10)*

U+2756

Tofu



⊑ *Asian — Individual Asian food inherits Asian (2026-10-10)*

U+1F217

PreservedEgg



⊑ *Asian — Individual Asian food inherits Asian (2026-10-10)*

U+1F218

CantoneseDuck



⊑ *Asian — Individual Asian food inherits Asian (2026-10-10)*

U+1F219

KelpFrond



⊑ *Sundry — Sundry item inherits Sundry (2026-10-10)*

U+328C

EucalyptusLeaf



⊑ *Sundry — Sundry item inherits Sundry (2026-10-10)*

U+328D

CandyBar



⊑ *Sundry — Sundry item inherits Sundry (2026-10-10)*

U+328E

LembasWafer



⊑ *Sundry — Sundry item inherits Sundry (2026-10-10)*

U+328F

CramRation



⊑ *Sundry — Sundry item inherits Sundry (2026-10-10)*

U+3290

FoodRation



⊑ *Sundry — Sundry item inherits Sundry (2026-10-10)*

U+32A9

KRation



⊑ *Sundry — Sundry item inherits Sundry (2026-10-10)*

U+32AA

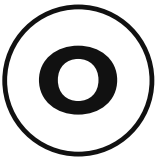
CRation



⊑ *Sundry — Sundry item inherits Sundry (2026-10-10)*

U+32AB

Sundry



⊑ *Deli — Sundry is part of Deli (2026-10-10)*

U+3267

CloveOfGarlic



⊑ *Sundry — Sundry item inherits Sundry (2026-10-10)*

U+329F

SprigOfWolfsbane



⊑ *Sundry — Sundry item inherits Sundry (2026-10-10)*

U+32A0

LumpOfRoyalJelly



⊑ *Sundry — Sundry item inherits Sundry (2026-10-10)*

U+32A2

Fruit



⊑ *Food — Fruit inherits Food — the fruit food category (2026-10-10)*

U+1F030

Apple



⊑ *Fruit — Fruit item inherits Fruit (2026-10-10)*

U+1F032

Orange



⊑ *Fruit — Fruit item inherits Fruit (2026-10-10)*

U+1F033

Pear



⊑ *Fruit — Fruit item inherits Fruit (2026-10-10)*

U+1F034

Melon



⊑ *Fruit — Fruit item inherits Fruit (2026-10-10)*

U+1F035

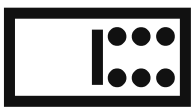
Banana



⊑ *Fruit — Fruit item inherits Fruit (2026-10-10)*

U+1F036

Carrot



⊑ *Fruit — Fruit item inherits Fruit (2026-10-10)*

U+1F037

Fig



⊑ *Fruit — Fruit item inherits Fruit (2026-10-10)*

U+1F038

Soup



⊑ *Tin — Soup inherits Tin — the soup category (2026-10-11)*

U+2473

Lemon



⊑ *Fruit — Fruit item inherits Fruit (2026-10-10)*

U+1F039

Lime



⊑ *Fruit — Fruit item inherits Fruit (2026-10-10)*

U+1F03A

Grapes



⊑ *Fruit — Fruit item inherits Fruit (2026-10-10)*

U+1F03B

Goodbar



⊑ *Candy — Goodbar inherits Candy (2026-10-11)*

U+1F000

Honee



⊑ *Candy — Honee inherits Candy (2026-10-11)*

U+1F001

DRation



⊑ *Candy — DRation inherits Candy (2026-10-11)*

U+1F002

Candy



⊑ *Food — Candy inherits Food — the candy category (2026-10-11)*

U+1F02A

Tomb



⊑ *Sundry — Tomb inherits Sundry*
(2026-10-11)

U+3248

Pesto



⊑ *Sundry — Pesto inherits Sundry*
(2026-10-11)

U+3249

DoveBar



⊑ *Candy — DoveBar inherits Candy*
(2026-10-11)

U+1F003

CrowBar



⊑ *Candy — CrowBar inherits Candy*
(2026-10-11)

U+1F029

Egg

Ø

⊑ *Creature — Egg inherits Creature (2026-10-11)*

U+00F8

Spinach



⊑ *Soup — Spinach (NetHack tin filling, Popeye strength) inherits Soup (2026-10-11)*

U+277C

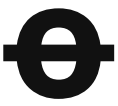
Meat



⊑ *Creature — Meat inherits Creature — the meat category (2026-10-11)*

U+4E3D

Creature



⊑ *Tin — Creature inherits Tin — the creature-meat tin category; the category that shuffles monsters as consumables (tins of monster meat) (2026-10-11)*

U+A74B

IntegerBounded

Ī

⊑ *Integer — IntegerBounded inherits Integer (new 2026-10-15)*

U+012A

IntegerBounded.max

⁇

⊑ *Integer — IntegerBounded.max is an Integer (new 2026-10-15)*

U+2047

IntegerBounded.min

⁇

⊑ *Integer — IntegerBounded.min is an Integer (new 2026-10-15)*

U+2047

IntegerBounded.epsilon

⁇

⊑ *Integer — IntegerBounded.epsilon is an Integer (new 2026-10-15)*

U+2047

IntegerBounded.nabla

⁇

IntegerBounded.nabla enforces bounded gets and sets (new 2026-10-15)

U+2047

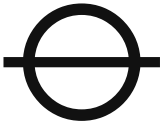
IntegerBounded.nabla

⁇

⊑ *Interpreter — IntegerBounded.nabla is an Interpreter (new 2026-10-15)*

U+2047

Angle



⊑ *IntegerBounded — Angle inherits IntegerBounded (new 2026-10-15)*

U+29B5

VirtualHorizontal



⊑ *Angle — VirtualHorizontal inherits Angle (new 2026-10-15)*

U+29D3

Egg

Ø

Egg stays empty till use — populated by the Creature monster-shuffle at runtime (2026-10-11)

U+00F8

VirtualDistance



⊑ *IntegerBounded — VirtualDistance inherits IntegerBounded (new 2026-10-15)*

U+27F0

VirtualVertical



⊑ *Angle — VirtualVertical inherits Angle (new 2026-10-15)*

U+29D7

Meat



Meat stays empty till use — populated by the Creature monster-shuffle at runtime (2026-10-11)

U+4E3D

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